What Determines the Effectiveness of 3D Field of Junctions?

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Abstract

We study when the 3D Field of Junctions (FoJ) denoiser helps and when it fails. FoJ fits each patch of a volume with a junction of planar wedges and averages the overlapping fits; it requires no training data and is designed for low-SNR volumes. We find that the released implementation, at its default configuration, reduces to an independent fit per patch followed by averaging, with no inter-patch coupling. From this mechanism follow three requirements for FoJ to help, each verified experimentally on CT reconstruction and three new denoising domains. First, the input must be unbiased: each region intensity is a mean of the voxels it covers, so a systematic offset in the input is reproduced rather than removed. Second, the structure must be piecewise-flat at or above the patch scale; features smaller than the patch cannot be represented by wedges, and the output collapses toward the mean. Third, the input noise must be high enough that the wedge model's bias is paid for by the noise it removes.

1 Introduction

Volumetric imaging modalities operate under dose and acquisition-time limits that keep the signal-to-noise ratio (SNR) low: CT scans limit the X-ray dose to reduce radiation exposure [1], lower-cost MRI systems use weaker magnetic fields [2], and electron tomography limits electron counts to avoid sample damage [3]. Denoising is therefore a routine step in volume analysis. The classical toolbox includes local filters such as Gaussian smoothing and median filtering; variational methods such as total-variation (TV) minimization [4, 5], which favor piecewise-constant solutions; and patch-recurrence methods such as non-local means [6] and BM3D [7]. Learned denoisers trained on paired data [8] currently dominate at moderate noise levels.

Field of Junctions (FoJ) [9] is a training-free structural prior for image denoising. It models each small image patch as a few constant-intensity regions separated by straight boundaries that meet at a vertex, called a junction, and produces a denoised image by averaging the overlapping patch models. 3D FoJ [10] extends the model to volumes: three planes meeting at a common vertex divide each cubic patch into constant-valued wedges. Because the wedge model captures sharp surfaces and corners rather than smooth content, it is designed for the low-SNR regime; the authors demonstrate it on low-dose X-ray CT, cryo-electron tomography, and point-cloud denoising (Figure 1), and report that it outperforms the baselines evaluated in their study: classical regularization, untrained neural representations [11, 12], and a denoiser trained on noisy data alone [13, 14]. A learned 2D successor, Boundary Attention [15], chooses patch sizes adaptively.

These demonstrations show that FoJ *can* work well on low-SNR volumes, but they do not characterize the conditions under which it does. The method's main hyperparameter is the patch

*Equal contribution; order determined by coin flip.

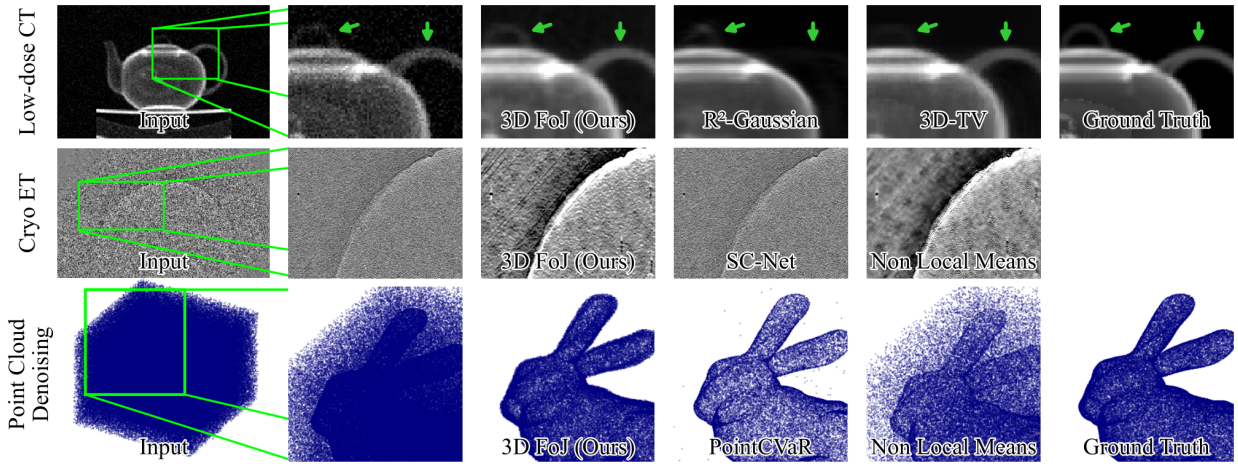


Fig. 1: Applications of 3D FoJ demonstrated in the original paper: low-dose CT (top), cryo-ET (middle), and point-cloud denoising (bottom). Figure reproduced from [10] (CC BY 4.0).

size R; the reported ablation finds reconstruction quality increasing monotonically with R on a low-dose CT task, and conjectures that the optimal patch size grows with noise level. Whether that holds, and more generally what determines when the method helps at all, is the subject of this report.

Our approach is to first establish what the algorithm actually computes at the configuration used for the published results (Section 3), and then to test the conditions that follow from it. Reproducing the CT ablation required re-implementing the simulation pipeline, since the public repository[1] releases the FoJ model but not the CT simulation, CGLS settings, view angles, or the M=3 region model used in the paper. We rebuild it following the R^2-Gaussian benchmark [16] whose data generator the paper used, and evaluate the released FoJ code on that task plus three denoising domains the paper did not test: BrainWeb MRI, grayscale micro-CT of Bentheim sandstone, and two binary porous-media volumes. Our implementation and experiment scripts are available in the accompanying repository.[2]

2 The 3D FoJ model

Let $y \in \mathbb{R}^{N^3}$ be a noisy volume and let $\{\Omega_p\}_p$ be cubic patches of side R on a grid with stride s. Each patch p carries parameters θ_p (three plane orientations and a shared vertex, nine numbers), which define soft indicator functions $u_{p,k}(\cdot;\theta_p)$ of M regions. Given θ_p, the region intensities are indicator-weighted means,

$$c_{p,k} = \frac{\sum_{x\in\Omega_p} u_{p,k}(x)\, y(x)}{\sum_{x\in\Omega_p} u_{p,k}(x)}, \tag{1}$$

[1] https://github.com/voilalab/3D-Field-of-Junctions
[2] https://github.com/samudraneel05/foj3d-factor-study

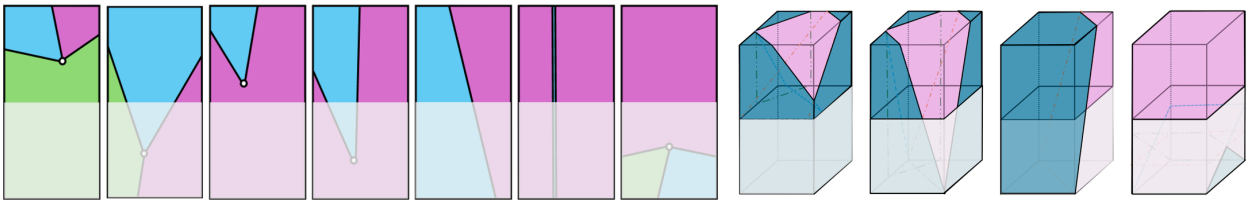


Fig. 2: Junction schematics, reproduced from [10] (CC BY 4.0). Left: 2D FoJ with M=3 regions; a vertex placed inside, on the boundary of, or outside the patch produces a three-way junction, a single edge, or a constant region, so the same model expresses all three local geometries. Right: the 3D FoJ variant, in which three planes meeting at a vertex divide the cubic patch into constant-valued wedges (M=2 shown).

and the patch rendering is $\hat{y}_p(x) = \sum_k u_{p,k}(x) c_{p,k}$. The full objective is

$$\mathcal{L}(\theta) = \sum_p \Big[\underbrace{\|\hat{y}_p - y|_{\Omega_p}\|_2^2}_{\text{data}} + \lambda_B \mathcal{C}_p^B(\theta) + \lambda_C \mathcal{C}_p^C(\theta) \Big], \tag{2}$$

where $\mathcal{C}^B, \mathcal{C}^C$ penalize disagreement between a patch's boundary and colour maps and the current global estimates. The output is the uniform average of overlapping renderings,

$$\hat{x}(x) = \frac{1}{|\{p : x \in \Omega_p\}|} \sum_{p:\, x \in \Omega_p} \hat{y}_p(x). \tag{3}$$

Optimization has two phases: N_{init} *initialization* iterations (coordinate descent over the nine parameters on a discrete grid, then a vertex line search along each plane normal) and N_{ref} *refinement* iterations (Adam on Eq. (2)). The default used for the paper's results is N_{init}=N_{ref}=1, R=10, s=2, M=3; the released code implements M=2, one convex wedge and its complement.

3 The effective estimator

Before examining when FoJ helps, we first establish what the algorithm computes at the configuration used for the published results, since the conditions under which the method can succeed follow from the estimator itself.

Equation (2) describes a coupled system: the boundary- and color-consistency terms tie each patch to the global estimates, so neighboring junctions are meant to agree on where surfaces lie and what intensities they take. That coupling is what makes the method a *field* of junctions rather than a collection of independent ones. At the configuration released by the authors and used for the published results, however, this coupling never engages, for two reasons. The consistency weights are ramped in only when the refinement stage runs for more than one iteration; the released schedule uses exactly one, so λ_B and λ_C remain zero throughout and the consistency terms contribute nothing. And the single refinement step that does run re-renders the volume from parameter values computed before the optimizer's update, so even it leaves the output unchanged.

What runs in practice is therefore simpler than the objective suggests: each patch is fitted independently,

$$\theta_p^\star = \arg\min_\theta \ \|\hat{y}_p(\theta) - y|_{\Omega_p}\|_2^2, \tag{4}$$

and the denoised volume is the average of the overlapping fits (Eq. (3)).

Table 1: Reported results for the ablation we reproduce (engine, P50, M=3, batch 6, stride 2; Table 3 of [10]).

R	4	6	8	10	12
PSNR (dB)	11.33	14.37	16.18	17.17	17.70
MS-SSIM	0.762	0.784	0.796	0.798	0.800
Runtime (min, RTX A6000)	7.2	10.5	13.1	17.2	24.7

This description is directly checkable. If patches never interact, the output cannot depend on how they are grouped: processing the volume in spatial chunks, each padded so that every patch sees its full support, reproduces the global implementation to $\sim 3 \times 10^{-7}$ on a CGLS reconstruction crop, and switching the refinement step on or off changes the output by at most 3.6×10^{-7}. The paper's own ablations are consistent with the same picture: batch sizes 1–6 give identical results, and additional refinement iterations, which are what activate the consistency terms, degrade rather than improve quality. All full-volume results reported below are within 0.01 dB of an exact global run.

The consequences for effectiveness are immediate. An estimator that only replaces each wedge of the input with a regional mean has no way to remove a *bias* already present in that input: the mean reproduces the offset rather than correcting it. It can only represent structure that is *piecewise-flat at the patch scale*: smooth texture and features smaller than R are averaged away. And its benefit is bounded by its bias: the wedge model is a coarse approximation, so it helps only where the noise it removes exceeds the error it introduces. The experiments of Section 5 test each of these conditions in turn.

4 Experimental setup

CT reconstruction. We reconstruct the 256^3 *engine* volume from 20 cone-beam views corrupted by Poisson noise, then apply FoJ, following the ablation of Table 3 in [10] (reproduced here as Table 1). The view angles, CGLS settings, and any preprocessing of the reconstruction are not specified in the paper, so we fix: the R^2-Gaussian scanner geometry (cone beam, source–detector distance 7, source–object distance 5, 256×256 detector); twenty stratified-jitter angles in $[0, 2\pi)$, seed 0; Poisson noise as in TIGRE's [17] `CTnoise` with P=50 expected photons per pixel in air ("P50"), defined relative to the maximum projection value; and CGLS [18] with 60 iterations. FoJ runs at M=2, N_{init}=N_{ref}=1, stride 2, $R \in \{4, 6, 8, 10, 12\}$, executed in halo-padded chunks on a single A100; one 256^3 run takes 225–325 s. Metrics are full-volume 3D PSNR (peak 1) and MS-SSIM [19] averaged over 360 forward projections.

Volume denoising. We run direct denoising on three domains with public ground truth: a 192^3 crop of Bentheim sandstone micro-CT (5.85 μm voxels, grains tens of voxels across [20]) under additive Gaussian noise at $\sigma \in \{15\%, 30\%, 40\%, 55\%\}$ of the dynamic range; the BrainWeb T1 phantom ($181 \times 217 \times 181$ [21]) under Rician noise at $\sigma \in \{3\% \ldots 13\%\}$; and two binary porous-media volumes (a synthetic sphere pack and a segmented Berea scan [22]) with feature sizes of about 2 voxels, under Gaussian noise at σ=15%. Baselines are Gaussian filtering, a 3^3 median filter, and TV [4, 5] at three weights, all under identical conditions; FoJ runs at $R \in \{6, 8, 12\}$, stride 2. We also report a boundary F-score over gradient-magnitude edge voxels, since edge preservation is FoJ's distinguishing feature.

Table 2: Engine 256^3, 20 jittered views, P50, TIGRE CGLS-60, released FoJ (M=2), stride 2. CGLS input PSNR: 6.92 dB unclamped, 9.97 dB clamped; input MS-SSIM 0.682. "–": run lost to a VM preemption.

R	4	6	8	10	12
Reported (M=3), PSNR	11.33	14.37	16.18	17.17	17.70
Ours, unclamped input, PSNR	14.72	16.93	17.99	18.58	**18.86**
Ours, unclamped input, MS-SSIM	0.786	0.829	0.854	0.870	**0.880**
Ours, clamped input, PSNR	–	14.24	14.48	14.58	**14.61**
Ours, clamped input, MS-SSIM	–	0.723	0.728	**0.728**	0.726
Gap ours (unclamped) − reported, PSNR	+3.39	+2.56	+1.81	+1.41	+1.16

5 Experiments

5.1 Input bias: clamping the reconstruction

By Eq. (1), every region intensity is a mean of the input voxels it covers, so a systematic offset in the input passes straight through the model. The natural source of bias in the CT pipeline is clamping: CGLS noise in the near-zero background is roughly symmetric, and clipping the reconstruction to $[0, 1]$ deletes the negative half, leaving a positive offset.

Table 2 shows the effect. Clamping raises the *input* PSNR by 3 dB but lowers the FoJ output by 2.7 dB (R=6) to 4.3 dB (R=12), and almost flattens the patch-size trend (0.37 dB between R=6 and R=12). Larger patches average more background voxels into each region estimate, so they lose the most from the biased input. The reported values lie between our clamped and unclamped rows, so the authors' preprocessing choice cannot be inferred.

With an unclamped input, the qualitative trend reproduces: PSNR and MS-SSIM increase monotonically with R, R=12 is best, and the marginal gains shrink (+2.2, +1.1, +0.6, +0.3 dB versus the reported +3.0, +1.8, +1.0, +0.5 dB). Our absolute values are 1.2–3.4 dB higher; the region count (M=2 versus M=3) accounts for at most about 0.14 dB by the reported region ablation.

5.2 Structure compatibility

The wedge model expresses flat regions with hard boundaries at roughly the patch scale or larger. Three tests probe this requirement, one on the input's artifact structure and two on the volume's own structure.

Artifact structure: the patch-size ordering reverses. Re-running the ablation with a simplified pure-PyTorch cone-beam projector (trilinear sampling along rays) in place of TIGRE changes the ordering. The simplified projector scales line integrals by 2.41× and rotates about a different volume axis, so the noise model, defined relative to the maximum projection value, produces artifacts of different shape and level at matched nominal noise. With CGLS-15 and P50, its input reaches 12.20 dB clamped (9.38 dB unclamped), comparable to the TIGRE input, yet R=6 or R=8 is now best and R=12 is worst (Table 3). A CGLS-iteration sweep never recovers a monotone trend. Two inputs of similar PSNR thus yield opposite hyperparameter conclusions: FoJ responds to the structure of the artifacts, not their energy. Consistent with this, a crop sweep on the R^2-Gaussian

Table 3: The same FoJ settings on CGLS-15 inputs from the simplified projector (engine 256^3, P50). The ordering is the opposite of Table 2.

R	4	6	8	10	12
Clamped input, PSNR	17.53	**17.84**	17.77	17.64	17.48
Unclamped input, PSNR	20.70	22.22	**22.28**	22.11	21.74

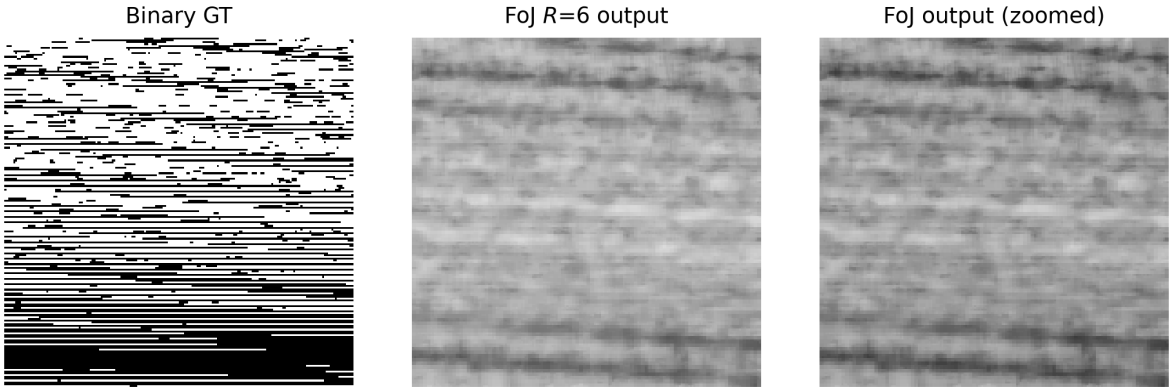


Fig. 3: Collapse on sub-patch-scale binary structure. Left: GT mid-slice (features about 2 voxels). Centre/right: FoJ R=6 output; only the coarsest layer structure survives and fine texture is averaged away.

foot and *jaw* volumes with the same projector puts the best R at 6–8 for engine and foot but 10–12 for jaw at every noise level; in these runs object scale, not noise level, sets the optimal patch size.

Smooth texture: MRI fails at every noise level. On BrainWeb T1 under Rician noise, FoJ never reaches the classical baselines. At σ=3% it trails a 3^3 median filter by 5.1 dB; at σ=13% the gap shrinks to 1.0 dB but never closes, and the ordering is R=6 > 8 > 12: larger patches quantize away more of the tissue texture that the wedge model cannot represent. The same holds when FoJ serves as the denoiser in a plug-and-play loop for accelerated MRI; at 4× undersampling it reaches 28.0 dB against 30.4 dB for TV-PnP.

Sub-patch-scale structure: total collapse. On the two binary volumes, features are about 2 voxels across, far below any tested patch size. FoJ collapses toward the global mean: output standard deviation 0.09 against 0.49 for the input, and 7.1 dB against the noisy input's own 19.5 dB. When an R-voxel patch contains only unmodelable speckle, its best fit is a constant (Figure 3), and averaging constants returns a constant.

5.3 Noise regime: the crossover

If FoJ carries a large approximation bias, it should lose whenever classical filters can recover the signal from local statistics alone, and win only where those filters break down. Bentheim micro-CT provides the compatible-structure test case: grains and pores are near-constant regions tens of voxels across.

Table 4 and Figure 5 show a clean crossover. FoJ trails at $\sigma \leq 30\%$, overtakes every classical baseline between 30% and 40%, and its margin grows with noise. The advantage is concentrated at

σ	Noisy	Gaussian	Median	TV (best w)	FoJ (best R)
15%	16.89	**28.51**	27.01	27.65	26.55
30%	12.00	25.26	22.26	**26.03**	25.64
40%	10.28	23.39	20.04	23.32	**24.93**
55%	8.76	21.05	17.60	16.30	**24.29**

Table 4: PSNR (dB) on the Bentheim volume. FoJ overtakes all classical baselines once the input PSNR falls below ∼11 dB (+1.5 dB at σ=40%, +3.2 dB at σ=55%) while trailing at moderate noise.

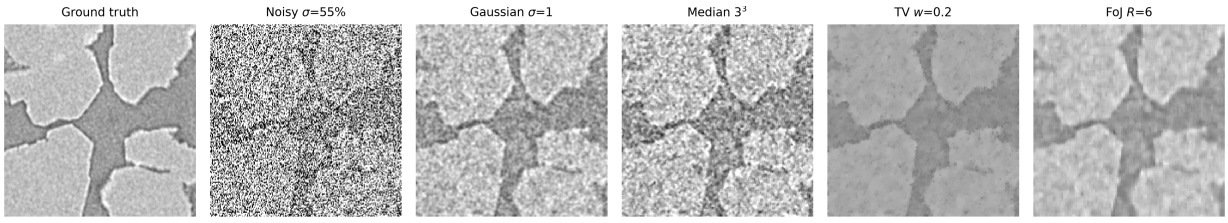


Fig. 4: Bentheim micro-CT, mid-slice, σ=55% Gaussian noise. FoJ (R=6) preserves the sharp grain–pore interfaces that Gaussian, median, and TV filtering blur.

boundaries: at σ=55% FoJ retains boundary-F1 = 0.85–0.88 against 0.54 for the best TV weight, and Figure 4 shows the grain–pore interfaces preserved where TV loses them. The ordering at σ=55% is R=8 > 6 > 12.

The same concentration appears inside the published results. The per-dataset table in [10] shows FoJ with the best PSNR on three of the five CT volumes at its lowest noise level (P50), on none of them at the next level (P100), and losing at every level on two volumes; the accompanying text likewise describes the higher-SNR results as mixed. The measured advantage of FoJ in the published data is thus confined to the same regime our experiments identify.

Figure 6 places all the denoising results on one axis: FoJ's margin over the best classical filter versus input PSNR. On compatible structure the margin grows monotonically with noise and crosses zero at an input PSNR of ≈11 dB, which is also the operating regime of the reported CT experiments: their inputs are ∼10 dB CGLS volumes. MRI stays negative throughout, because its violation is structural rather than noise-driven. On volumetric data with phase-scale structure and input PSNR $\lesssim$ 11 dB, the released 3D FoJ implementation outperforms Gaussian, median, and TV denoising by up to 3.2 dB.

6 When does FoJ help?

Each result above follows from the separable estimator of Section 3 rather than standing alone (Table 5). The method succeeds when three conditions hold together: the input is unbiased, the structure is piecewise-flat at or above the patch scale, and the noise is high enough that the wedge model's bias is paid for by the noise it removes. The micro-CT setting is the one case in our study where all three hold.

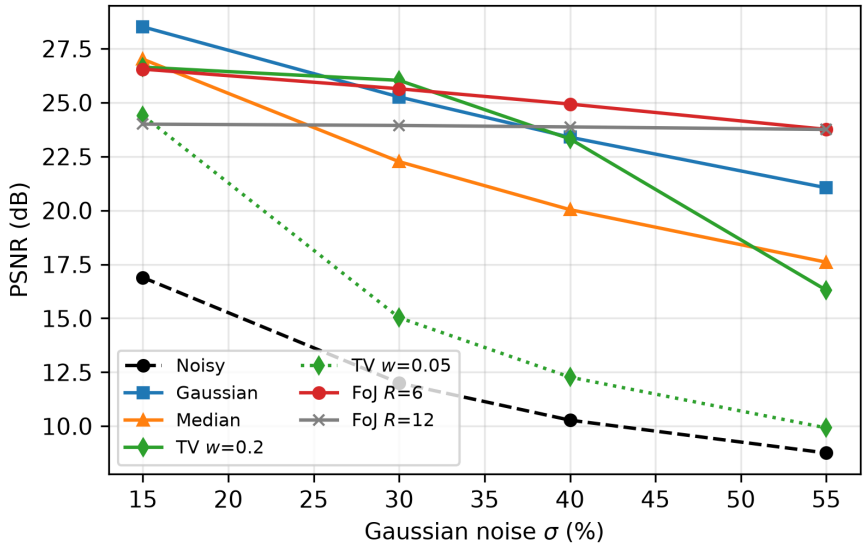


Fig. 5: PSNR versus noise level on Bentheim micro-CT. FoJ (red) overtakes the classical filters between σ=30% and 40%.

7 Conclusion

At its default configuration, 3D Field of Junctions reduces to an independent per-patch wedge fit followed by overlap averaging. Its regime of validity follows from that mechanism. The input must be unbiased: clamping a reconstruction to $[0, 1]$ costs up to 4.3 dB and nearly eliminates the patch-size trend. The structure must be piecewise-flat at or above the patch scale: smooth MRI tissue fails at every noise level, and sub-patch-scale features collapse the output to the mean. The input SNR must be low enough that the wedge model's approximation bias is paid for by the noise it removes: on Bentheim sandstone micro-CT, FoJ overtakes Gaussian, median, and TV denoisers below an input PSNR of about 11 dB and leads by up to 3.2 dB. The reported patch-size trend reproduces under a faithful forward model but reverses under a different projector, so hyperparameter conclusions for structural priors depend on simulation fidelity. Outside the low-SNR, flat-structure regime, simple classical filters are the better choice.

Future work. (1) Dose sweeps (P50/P100/P1000) to test whether the optimal patch size grows with noise level; (2) additional objects to separate object scale from noise; (3) FoJ inside a proximal-gradient loop, as in the paper's main CT experiment; (4) BM4D and non-local-means baselines; (5) real scanner noise rather than synthetic corruption; (6) junction-shaped primitives inside the reconstruction rather than after it.

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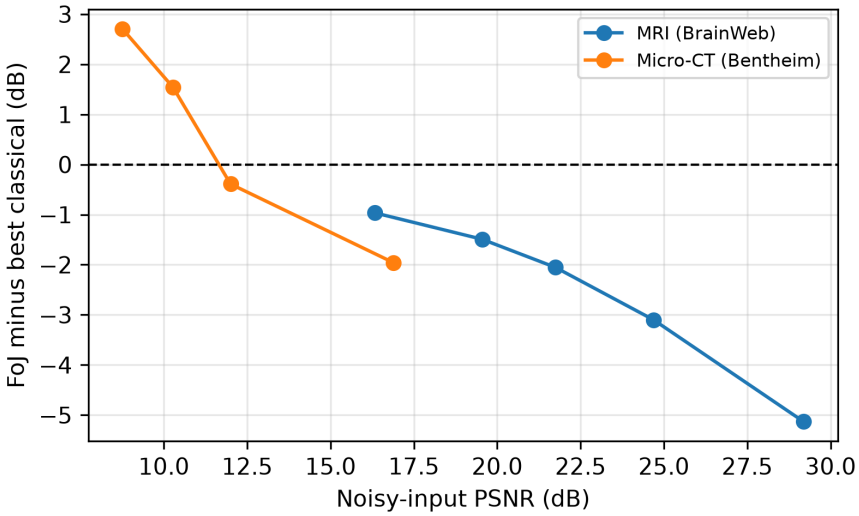


Fig. 6: FoJ PSNR margin over the best classical denoiser versus input PSNR. Above ≈11 dB input, classical filters are better; below it, FoJ is better on piecewise-constant volumes.

Table 5: The three requirements and the experiment that tested each.

Requirement	Evidence	Rule of thumb
Unbiased input	Clamping CGLS output to $[0, 1]$ costs 2.7–4.3 dB and flattens the patch-size trend.	Feed FoJ an unclamped, bias-free input.
Compatible structure	Simplified projector flips the patch ordering at matched PSNR; smooth MRI tissue trails a median filter by 1.0–5.1 dB; sub-patch binary features collapse FoJ to the mean (7.1 dB).	Regions flat and larger than the patch; check both GT structure and artifact structure.
Low input SNR	Micro-CT crossover at ≈11 dB input; FoJ leads by +1.5 to +3.2 dB below it.	Use FoJ where classical filters fail, not where they work.

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